DSAA3073 · THEORIES OF DATA SCIENCE

OLS Beyond Memorization

Understanding the Normal Equations

Why does $X^\top$ appear in least squares?

$$X^\top \left(y - X\hat{\beta} \right) = 0$$

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Outline

1. The problem and its spaces
2. Projection and optimization
3. Gram matrices and the adjoint
4. Rank, stability and ridge

Start from the least-squares problem

Why do we left-multiply by $X^\top$? What makes this choice meaningful?

Observed data and a linear model

$$X \in \mathbb{R}^{n \times d}, \quad y \in \mathbb{R}^n$$

A model may write $y = X\beta_* + \varepsilon$.
The noise ε need not be orthogonal to the feature columns.

The fitting problem

$$\hat{\beta} \in \operatorname*{argmin}_{\beta} \|y - X\beta\|^2$$

Choose coefficients to minimize squared residuals.
The fitted residual is $r = y - X\hat{\beta}$.



The normal equations characterize a least-squares fit. They do not follow by simply dropping the noise term.

1 · Dimensions locate the transpose

Object	Space / size	Meaning
β	$\mathbb{R}^d$	One coefficient per feature
$X\beta$ and y	$\mathbb{R}^n$	One prediction or response per sample
X	$n \times d$	Maps coefficients to predictions
$X^\top$	$d \times n$	Maps a data vector to feature inner products

$$\beta \xrightarrow{X} X\beta \qquad r \xrightarrow{X^\top} X^\top r$$



Dimensions explain where $X^\top r$ lives. They do not yet explain why its value must be zero at the fit.

A running example: fit one constant

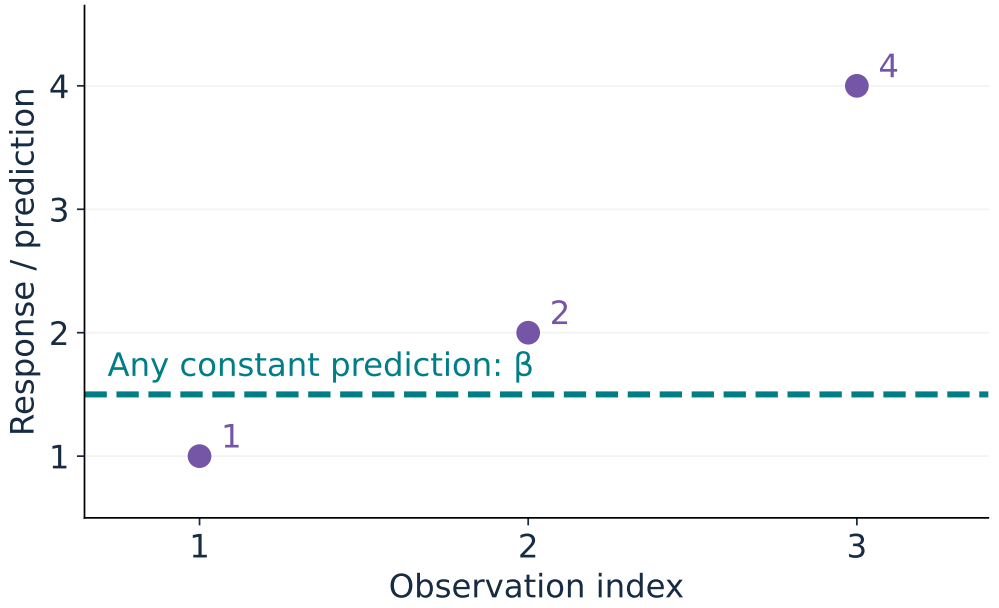
We observe three responses and predict the same value for each:

$$X = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad y = \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}, \quad \beta \in \mathbb{R}.$$

$$L(\beta) = (1-\beta)^2 + (2-\beta)^2 + (4-\beta)^2.$$

Which constant gives the smallest squared error?

One coefficient, three residuals



💡 The feasible predictions form a line in $\mathbb{R}^3$: every vector of the form $(\beta, \beta, \beta)^\top$.

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2 · Least squares is orthogonal projection

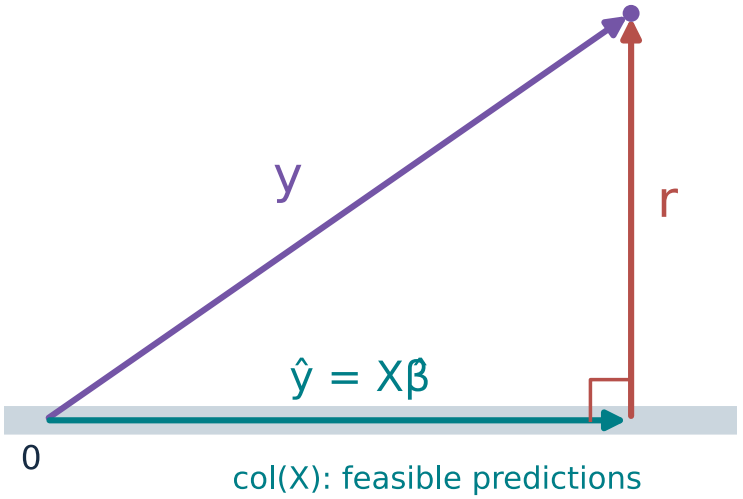
All predictions lie in $\mathrm{col}(X)$.
At the closest prediction $\hat{y} = X\hat{\beta}$, $r = y - \hat{y}$ is perpendicular to this entire subspace.

$$X^\top r = 0$$

$$X^\top X\hat{\beta} = X^\top y$$

"Normal" here means perpendicular.

Schematic view in a two-dimensional plane



💡 $y = \hat{y} + r$: the fitted part lies in $\mathrm{col}(X)$; the unexplained part is orthogonal to it.

Why orthogonality guarantees the minimum

Suppose $X^\top r = 0$, with $r = y - X\hat{\beta}$. Move to any coefficients $\hat{\beta} + h$.

Compare the losses

$$\begin{aligned} L\left(\hat{\beta} + h\right) &= \|r - Xh\|^2 \\ &= \|r\|^2 - 2r^\top Xh + \|Xh\|^2 \\ &= L\left(\hat{\beta}\right) + \|Xh\|^2 \geq L\left(\hat{\beta}\right). \end{aligned}$$

The cross term vanishes because $r^\top Xh = \left(X^\top r\right)^\top h = 0$.



Every feasible change can only increase the loss or leave it unchanged. Full column rank makes every nonzero coefficient change increase it.

3 · The gradient yields the same condition

Differentiate the squared loss

$$L(\beta) = (y - X\beta)^\top (y - X\beta)$$

$$\nabla L(\beta) = -2X^\top (y - X\beta)$$

At a stationary point:

$$\nabla L\left(\hat{\beta}\right) = 0$$

$$X^\top X\hat{\beta} = X^\top y.$$

Why this is a global minimum

The Hessian is positive semidefinite:

$$\nabla^2 L = 2X^\top X,$$

$$h^\top \nabla^2 L h = 2\|Xh\|^2 \geq 0.$$

The loss is convex. With full column rank, it is strictly convex.



Setting the gradient to zero solves the optimality condition. It is not generally one gradient-descent step.

Check the fit in the same example

Normal equation and residual

$$X^\top X = 3, \quad X^\top y = 7$$

$$\hat{\beta} = \frac{7}{3}$$

$$r = \begin{pmatrix} -\frac{4}{3} \\ -\frac{1}{3} \\ \frac{5}{3} \end{pmatrix}$$

$$X^\top r = -\frac{4}{3} - \frac{1}{3} + \frac{5}{3} = 0$$

Gradient and loss

$$L'(\beta) = 6\beta - 14$$

$$L'\left(\frac{7}{3}\right) = 0$$

$$L\left(\frac{7}{3}\right) = \frac{14}{3}$$

$$L\left(\frac{7}{3} + \delta\right) = \frac{14}{3} + 3\delta^2$$



The mean, the projection, and the zero-gradient solution agree. Here the residuals sum to zero because X contains the constant feature.

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4 · The Gram matrix records feature geometry

Let X_j be column j of X. The Gram matrix $G = X^\top X$ satisfies

$$G_{jk} = X_j^\top X_k = \sum_{i=1}^{n} X_{ij} X_{ik}.$$

Its entries

- $G_{jj} = \|X_j\|^2$: feature magnitude.
- G_{jk}: alignment between two feature columns.
- G is symmetric and positive semidefinite.

Its action

$$h^\top G h = \|Xh\|^2$$

This measures how much a coefficient change alters the predictions.
G is invertible exactly when $\operatorname{rank}(X) = d$.



The normal equations are square, but their significance comes from feature geometry. Square does not imply invertible.

5 · Match the feature inner products

Each normal equation matches one observed inner product with its fitted counterpart.

$$X_j^\top y = X_j^\top X\hat{\beta} = \sum_{k=1}^{d}\left(X_j^\top X_k\right)\hat{\beta}_k$$

Quantity	Interpretation
$X_j^\top y$	Alignment of feature j with the observed response
$X_j^\top X_k$	Alignment between features j and k
$X_j^\top r = 0$	No residual component along feature j



Inner products depend on scale and centering. They are not automatically Pearson correlations or individual causal contributions.

Making a square system is not enough

For the constant-fit example, choose a different 1×3 map $A = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix}$.

Arbitrary compression

$$AX\beta = Ay$$
$$\beta = 1$$

This square system has a unique solution, but

$$L(1) = 0^2 + 1^2 + 3^2 = 10.$$

Least-squares compression

$$X^\top X\beta = X^\top y$$
$$\beta = \frac{7}{3}$$

The residual is orthogonal to the model space, and

$$L\left(\frac{7}{3}\right) = \frac{14}{3} < 10.$$



Many maps have the right dimensions. $X^\top$ is special because it tests the residual against every feasible prediction direction.

6 · The adjoint explains why the transpose

Under the standard Euclidean inner products, the adjoint of X is $X^\top$:

$$\langle Xh, r\rangle_{\mathbb{R}^n} = \langle h, X^\top r\rangle_{\mathbb{R}^d}.$$

A direction in each space

A parameter direction h produces a prediction direction Xh.
Pairing the residual with Xh gives exactly the same scalar as pairing $X^\top r$ with h.

The optimality condition

$$X^\top r = 0$$
$$\leftrightarrow$$
$$\langle Xh, r\rangle = 0 \quad \forall h$$

Thus the residual is orthogonal to $\mathrm{col}(X)$.



The transpose preserves the pairing between parameter and data directions. The choice of inner products matters.

Six views, one optimality condition

View	What it explains
Dimensions	$X^\top r$ lives in parameter space $\mathbb{R}^d$.
Geometry	r is perpendicular to all of $\mathrm{col}(X)$.
Optimization	$\nabla L\left(\hat{\beta}\right) = -2X^\top r = 0.$
Gram matrix	$X^\top X$ encodes how coefficient changes affect predictions.
Inner products	$X_j^\top y = X_j^\top \hat{y}$ for every feature j.
Adjoint	$\langle Xh, r\rangle = \langle h, X^\top r\rangle$.



To explain the normal equations, connect the spaces, the loss, and the residual. Each view answers a different part of the same question.

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Rank controls coefficient uniqueness

Full column rank: $\operatorname{rank}(X) = d$

$X^\top X$ is positive definite.
There is one coefficient vector:

$$\hat{\beta} = (X^\top X)^{-1} X^\top y.$$

This formula requires full column rank.

Rank deficient: $\operatorname{rank}(X) < d$

$X^\top X$ is singular, but least squares still has minimizers.
If $Xh = 0$, then $\hat{\beta} + h$ produces the same fitted values as $\hat{\beta}$.
The fitted vector $\hat{y}$ remains unique.

Example: duplicate the constant feature, $X = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{pmatrix}$. Every pair with $\beta_1 + \beta_2 = \frac{7}{3}$ gives the same fit to $(1, 2, 4)^\top$.



Different coefficient vectors can represent the same unique projection. Identifiability of parameters is a separate question.

Separate coefficients from fitted values

Operator	Map	What it returns
X^+	$\mathbb{R}^n \to \mathbb{R}^d$	Minimum-norm least-squares coefficients
$P_X = XX^+$	$\mathbb{R}^n \to \mathbb{R}^n$	Orthogonal projection onto $\mathrm{col}(X)$

$$\hat{\beta}_{\min} = X^+ y$$

With full column rank only:

$$X^+ = (X^\top X)^{-1} X^\top.$$

$$\hat{y} = P_X y$$

The projector is symmetric and idempotent:

$$P_X^\top = P_X, \quad P_X^2 = P_X.$$



X^+ returns coefficients; XX^+ returns predictions. These operators have different domains and meanings.

Small singular values amplify response noise

Use the compact SVD $X = U_q \Sigma_q V_q^\top$, with $q = \operatorname{rank}(X)$ and $\sigma_i > 0$.

$$X^+ y = \sum_{i=1}^{q} \frac{\langle u_i, y \rangle}{\sigma_i} v_i.$$

Perturb one data direction

If $\Delta y = \delta u_i$, then

$$\Delta \hat{\beta}_{\min} = \frac{\delta}{\sigma_i} v_i.$$

Small σ_i means large coefficient sensitivity.

A numerical scale

Let $\sigma_i = 0.01$ and $\delta = 0.02$.

$$\left\| \Delta \hat{\beta}_{\min} \right\| = 2.$$

Yet $\Delta \hat{y} = \delta u_i$: fitted training values need not change greatly.



Coefficient instability is not the same as instability of fitted training values. Small singular directions reveal the distinction.

Ridge limits amplification in weak directions

Penalize all coefficients, with $\lambda > 0$:

$$\hat{\beta}_\lambda = \operatorname*{argmin}_\beta (\|y - X\beta\|^2 + \lambda\|\beta\|^2).$$

The modified normal equations

$$(X^\top X + \lambda I)\hat{\beta}_\lambda = X^\top y$$

The compact SVD form

$$\hat{\beta}_\lambda = \sum_{i=1}^{q} \frac{\sigma_i}{\sigma_i^2 + \lambda} \langle u_i, y \rangle v_i$$

The same perturbation

For $\sigma_i = 0.01$, $\delta = 0.02$, and $\lambda = 0.01$:
OLS coefficient change: 2.
Ridge coefficient change:

$$\frac{0.01}{0.01^2 + 0.01} \cdot 0.02 \approx 0.0198.$$



Ridge trades shrinkage bias for reduced sensitivity. The penalty makes the coefficient solution unique even when X is rank deficient.

Ridge fits are shrunk, not orthogonally projected

The fitted-value operator is

$$S_\lambda = X(X^\top X + \lambda I)^{-1} X^\top = U_q \operatorname{diag}\left(\frac{\sigma_i^2}{\sigma_i^2 + \lambda}\right) U_q^\top.$$

Orthogonal projection

P_X keeps each direction in $\operatorname{col}(X)$ with weight 1.

$$P_X^2 = P_X$$

Applying it again changes nothing.

Ridge smoothing

S_λ uses weights strictly between 0 and 1 on nonzero singular directions.

For nonzero X, $S_\lambda^2 \neq S_\lambda$.

Applying it again shrinks further.



The ridge residual satisfies $X^\top\left(y - X\hat{\beta}_\lambda\right) = \lambda\hat{\beta}_\lambda$. It is generally not orthogonal to $\operatorname{col}(X)$.

Read the two kinds of spectral weights

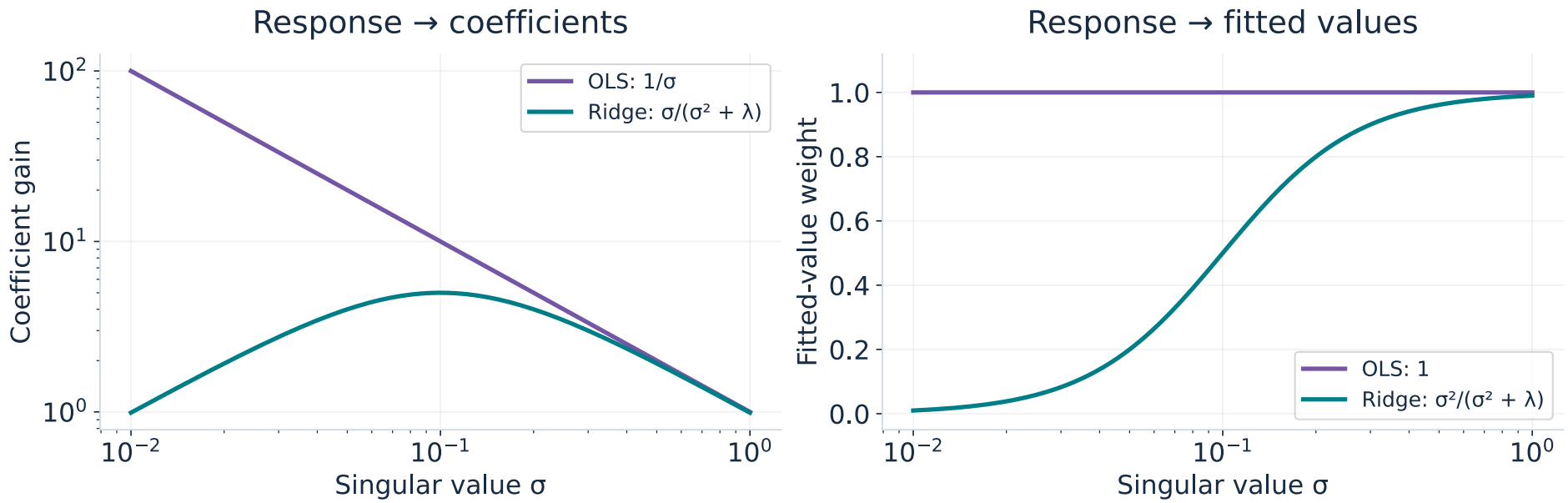


Illustration: $\lambda = 0.01$. Left: how data components affect coefficients. Right: how much each fitted component is retained.



For numerical computation, use QR or SVD-based solvers. Forming $X^\top X$ squares the 2-norm condition number when X has full column rank.

Explain the equations without memorizing them

Why $X^\top$?

It tests residuals against all feasible prediction directions, via the adjoint identity.

When is the inverse valid?

Exactly when X has full column rank. Otherwise use the pseudoinverse or describe all minimizers.

What is being projected?

y is projected to $\hat{y} = XX^+y$ in data space; coefficients live in parameter space.

What changes under ridge?

Weak directions receive smaller weights. The fit is a linear smoother, with a different residual condition.



Reconstruct $X^\top\left(y - X\hat{\beta}\right) = 0$ from the loss or from geometry. Then state the rank and regularization assumptions before using a formula.